

Smart Dustbin Using Arduino Uno

Correct Arduino Code for Working Prototype

Components Used

Component	Arduino Connection
HC-SR04 Ultrasonic Sensor	TRIG → D9, ECHO → D10, VCC → 5V, GND → GND
SG90 Servo Motor	Signal → D6, VCC → 5V, GND → GND
16×2 I2C LCD	SDA → A4, SCL → A5, VCC → 5V, GND → GND
Buzzer (Optional)	Positive → D7, Negative → GND

Arduino Code

```
#include <Servo.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// ----- PIN DEFINITIONS -----
#define TRIG_PIN 9
#define ECHO_PIN 10
#define SERVO_PIN 6
#define BUZZER_PIN 7

// ----- OBJECTS -----
Servo lidServo;

// Common I2C LCD address: 0x27
// If your LCD uses 0x3F, change 0x27 to 0x3F.
LiquidCrystal_I2C lcd(0x27, 16, 2);

// ----- SETTINGS -----
const int detectionDistance = 20; // Detection distance in cm
const int openAngle = 90; // Lid open position
const int closeAngle = 0; // Lid closed position
const unsigned long openTime = 3000; // Lid open time in milliseconds

void setup()
{
  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);

  pinMode(BUZZER_PIN, OUTPUT);
  digitalWrite(BUZZER_PIN, LOW);

  lidServo.attach(SERVO_PIN);
  lidServo.write(closeAngle);

  lcd.init();
  lcd.backlight();

  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Smart Dustbin");
  lcd.setCursor(0, 1);
  lcd.print("Status: CLOSED");

  delay(2000);
}

void loop()
{
  long duration;
  int distance;

  // Generate ultrasonic trigger pulse
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);

  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);

  digitalWrite(TRIG_PIN, LOW);

  // Measure echo time
```

```

    duration = pulseIn(ECHO_PIN, HIGH);

    // Calculate distance in centimeters
    distance = duration * 0.034 / 2;

    // Detect hand/object
    if (distance > 0 && distance <= detectionDistance)
    {
        openLid();
    }

    delay(100);
}

void openLid()
{
    lidServo.write(openAngle);

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Smart Dustbin");
    lcd.setCursor(0, 1);
    lcd.print("Status: OPEN");

    // Short buzzer beep
    digitalWrite(BUZZER_PIN, HIGH);
    delay(100);
    digitalWrite(BUZZER_PIN, LOW);

    delay(openTime);

    closeLid();
}

void closeLid()
{
    lidServo.write(closeAngle);

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Smart Dustbin");
    lcd.setCursor(0, 1);
    lcd.print("Status: CLOSED");
}

```

How the Prototype Works

The HC-SR04 detects a hand or object within 20 cm. Arduino Uno processes the measured distance and commands the SG90 servo to open the lid to 90°. The LCD displays the OPEN status and the optional buzzer gives a short beep. After 3 seconds, the servo returns to 0° and the LCD displays CLOSED.

Important Notes

1. The code assumes a 16×2 LCD with an I2C module at address 0x27. If the LCD does not display text, try 0x3F instead.
2. If the servo causes Arduino resets, power the SG90 from a suitable regulated 5V supply and connect its ground to Arduino GND.
3. Adjust openAngle and closeAngle to match the actual mechanical position of your dustbin lid.